1. Authentication, Onboarding & Access Control (AUTH)

- New Super Admin Onboarding: Public /register route. A 3-step wizard for new users to create their Account, Organization, and first Clinic/Workspace.
- Existing User Org Creation: Logged-in users (e.g., a Doctor) can click "+ Create New Organization" in the workspace switcher to provision a new clinic without creating a duplicate account.
- Staff Invitation Flow: Staff cannot self-register for a clinic. Admins invite them via email + role. Users set passwords via /accept-invite?token=xyz.
- Login / Logout: JWT access tokens (15m), refresh tokens (7d) in HttpOnly cookies.
- Dynamic Token Scoping: Backend endpoint /auth/switch-workspace to re-issue JWTs with different org_id, clinic_id, and role payloads based on the selected workspace.
- Rate Limiting & Lockout: 5 failed logins in 10 mins locks the account for 15 mins (Redis).
- Audit Trail: Record who, what, when, before/after values for all create/update/delete actions.

2. Multi-Tenancy & Hierarchy (MT)

- Global Identity Architecture: Separate the users table (email/password) from the user_memberships pivot table (mapping users to orgs/clinics with specific roles).
- DB Hierarchy: organizations → clinics (workspaces) → user_memberships / patients.
- Context Switcher UI: Dropdown for users to switch between multiple clinics they belong to. Shows their role per clinic (e.g., "Central Clinic - Doctor", "My Private Practice - Super Admin").
- Strict API Isolation: Backend Guard reads clinic_id and organization_id strictly from the JWT token.
- Tenant Isolation Test: Automated test proving User A cannot access Clinic B's patient data.

3. User Management & Roles (USR)

- User Directory: Table of staff for the active clinic, showing email, role, status, and last login.
- Cross-Tenant Role Management: An admin can assign a user multiple roles across different clinics.
- Role Assignment Matrix: 4 Roles (Super Admin, Clinic Admin, Doctor, Receptionist).
- Suspend/Revoke Access: Instantly block access to a specific clinic and invalidate tokens without deleting historical audit logs.

4. Hierarchical Settings (SET)

- Organization Settings (Super Admin): Global profile, Branches management, global cancellation policies, data retention rules.
- Clinic/Workspace Settings (Clinic Admin): Local address, timezone, local currency, departments, Rooms & Equipment management, operational hours, local appointment durations.
- User Settings (All Staff): Personal profile, language, notification channels, accessibility toggles.
- Unsaved Changes Warning: Prompt if navigating away with unsaved form data.

5. Internationalization (I18N)

- Language Switcher: English, Arabic, French.
- RTL Support: Full right-to-left layout mirroring for Arabic.
- Localized Data: Dates, times, and numbers formatted using Intl API.
- API Error Translation: Backend returns error codes; frontend translates them.
- Font Handling: Inter for EN/FR, Noto Sans Arabic for AR.

6. Patients (PAT)

- Patient Directory: Table with search, filters, and pagination.
- Patient Profile: Tabs for Personal Info, Upcoming Appointments, Past Visits, Clinical Notes.
- Field-Level Permissions: Receptionists see admin data; Doctors see clinical notes.
- Digital Intake Forms: Send forms (demographics, consent) pre-visit.
- Duplicate Detection: Warns if name + DOB or phone exists.

7. Doctors, Staff & Resources (DOC)

- Staff Directory: Card and Table view with utilization % and current status.
- Working Hours & Availability: Set weekly schedules, multiple intervals.
- Breaks, Leaves & Rostering: Block out time for lunch, holidays, and manage shift rosters for receptionists/nurses.
- Room/Equipment Assignment: Tie doctor availability to physical rooms to prevent double-booking spaces.
- Slot Generator: Backend function that calculates exact free slots.

8. Appointments & Calendar (APP)

- Appointment List & Calendar Views: Day, Week, Month views. Color-coded.
- 5-Step Booking Wizard: Select Patient → Doctor → Date/Time → Details → Review.
- State Machine: Strict lifecycle (Scheduled → Confirmed → Checked In → Completed / Cancelled / No-show).
- Conflict Detection: Postgres exclusion constraints + Redis slot locking (5-min TTL).
- Waitlist Auto-Fill: If an appointment is cancelled, automatically offer the slot to the waitlist.
- Reminders: BullMQ delayed jobs to send mock SMS/Email.

9. Waiting List & Queue (QUE)

- QR Code Self Check-in: Patients scan a QR code upon arrival to self-register into the queue.
- Priority Triage: Normal, Urgent, Emergency.
- "Call Next" Logic: Priority → Earliest appointment → Earliest check-in. Redis ZSET prevents race conditions.
- Staff Actions: Start consultation, complete, notify delay, mark no-show.
- Public Waiting Room Display (/display/:clinicId): Read-only screen showing queue number and room only.

10. Smart Scheduling (SMT)

- Metrics Service: Calculate utilization, idle gaps, estimated waiting time.
- Deterministic Optimizer: Greedy/local search algorithm to suggest moving appointments.
- Optimization Grid: Drag-and-drop appointment blocks.
- Insights Panel & Preview: Show suggested moves, expected wait time reduction. Preview before applying.
- Rollback & Stale Check: Undo within 15 mins. Reject apply if schedule changed since proposal generation.

11. Billing & Payments (Lightweight) (BIL)

- Auto-Generated Invoices: Generate invoice based on appointment type/services.
- Point-of-Care Payments: Accept and record payment (cash/card) at check-in/checkout.
- Receipt Generation: Generate printable/PDF receipt.
- Insurance Card Capture: Upload/record policy number for admin purposes.

12. Realtime - Chat & Notifications (RT)

- Socket.IO Integration: JWT auth on handshake. Rooms per clinic/user.
- Staff Chat: Direct and group chats. Typing indicators, read receipts.
- Notification Center: Bell icon with unread counter. Categories: Reminders, Cancellations, Conflicts.
- Postgres Persistence: Store chats and notifications in Postgres.

13. Analytics & Command Center (ANA)

- Executive Dashboard: 4 KPI cards, today's schedule, 4 charts.
- Live Floor Board (Command Center): Real-time heatmap of wait times per doctor, room occupancy, and patient states.
- Reports Page: Date range picker with comparison against previous period.
- Export: Streamed CSV (UTF-8 with BOM) and server-rendered PDF.

14. Optional AI Enhancements (Bonus)

- AI Quick Book: Natural language parsing to pre-fill the booking wizard.
- AI Message Drafter: Draft delay/cancellation SMS/Emails.
- AI Analytics Summarizer: Generate plain-text summary of the charts.